Learning Financial Jumps With Infinite Gaussian Mixture

Yuru Sun, Worapree (Ole) Maneesoonthorn, Yong Song, Wei Wei

Abstract

We propose a Bayesian nonparametric method using Dirichlet process mixture to model jumps. This new model encompasses existing parametric jump specification, but allows for extreme jumps to occur in either tail of the return distribution via an infinite mixture model. The flexibility in our model specification allows for multi-modal jump size distribution. We investigate the robustness of our proposed model under different jump intensity and size specifications under simulations and verify the practical use of our model with empirical data in jump detection, data-driven jump size density and predictive return distribution.

Keywords: stochastic volatility, Bayesian nonparametric, state space models, option pricing

1 Introduction

Modelling discontinuous and extreme price movements - jumps is crucial for risk management and option pricing in financial market. The stylized facts of financial return distribution and implied volatility(IV) smirk/smile motivate the usage of the stochastic volatility (SV) and jump-diffusion models. The SV models assume volatility to follow a stochastic process, and are proven to be an improvement from Black-Scholes Merton model (BSM) by Black and Scholes (1973) in option pricing literature (Heston, 1993; Hull and White, 1987; Scott, 1987; Wiggins, 1987; Fouque *et al.*, 2000). However, the SV model alone is inadequate in modelling extreme and unexpected price movements, since it only targets at the persistence of volatility (Eraker *et al.*, 2003; Gatheral, 2011). Depending on the assumption of innovation, it might not capture the fat tails of the return distribution very well, but improvements can be made by approximating the distribution of error term (Kim *et al.*, 1998; Jensen and Maheu, 2010).

To help accommodate fat tails of the return distribution and capture the discontinuous and extreme events simultaneously, while retaining their economic meaning, jump diffusion model is introduced as a complementary to the SV model. Previous literature show that the stochastic volatility jump (SVJ) model captures the return distribution better and generates better IV surface that fits to the empirical shape of IV, especially for short maturity options. See the work of Bates (1996); Cont (2001); Eraker *et al.* (2003); Eraker (2004); Gatheral (2011).

Conventionally, jumps are modelled by two components - jump arrival and jump size. The jump arrival follows a discontinuous point process and the jump size is assumed to follow a particular distribution. The assumption of jump size distribution is important for model performance and various types of distributions have been studied. Merton (1976) firstly proposes a jump diffusion model where a normal distribution is assumed. Even though this assumption is convenient for pricing options using characteristic function, it is found to be inadequate to capture the fat tails of return, especially during very volatile periods. Moreover, it does not generate enough skew in the shape of IV. Kou and Wang (2004); Kou (2007) propose to use double exponential or power-type distributions for jump size so that the model can cover the extreme fat tails and it is shown that adding jumps improves the estimation of tail distribution and provides better estimation of Value-at-Risk (VaR).

The dynamic of jump itself can be complicated. Various extensions to the SVJ model has been studied. Bates (2000) and Pan (2002) show that only considering price jumps is inadequate and prove the existence of volatility jumps. The clustering effect of price jump itself and between price and volatility jumps is also investigated. Duffie *et al.* (2000) and

Fulop *et al.* (2015) model the co-jumps effect of price and volatility.Eraker *et al.* (2003) show that the jumps are clustered, which breaks the assumptions about constant jump arrival rate (intensity) and independent and identically distributed jump sizes. Aït-Sahalia *et al.* (2015) introduce the mutually exciting jump process to continuous Brownian process and develop Hawkes jump-diffusion model to capture the price jump clustering.

On the other hand, the flexibility of modelling jumps is raised as well. Firstly, Chan and Maheu (2002) and Maheu and McCurdy (2004) look into the time-varying feature of jumps. They investigate the jump dynamics and propose a model where the jump intensity is allowed to change over time and jump size distribution is assumed to be normal with time-varying mean and variance. In addition,Maneesoonthorn *et al.* (2017) improves the flexibility of modeling jump size distribution by considering the fact that jump can occur in both positive and negative tails of return distribution. They model the direction of jumps separately as a Bernoulli distribution and assume jump magnitude is exponentially distributed, with the model validated empirically.

The literature has shown the importance and usefulness of jump models. Less accurate model specification of jumps can lead to less accurate model generated IV and return distribution, and hence poor risk management. Although parametric SVJ models bring convenience in estimation and option pricing, the assumption of jump size distribution is inflexible and can be misspecified.

Our paper proposes an innovative Bayesian nonparametric SVJ model where the assumption of jump size distribution is relaxed and the model is general, flexible and data-driven. We model the jump size distribution using Dirichlet process mixture (DPM) (Ferguson, 1973; Antoniak, 1974; Escobar and West, 1995). It is modeled as an infinite mixture of a particular distribution class and the weights of the mixture follows a stick-breaking process (Sethuraman, 1994). This allows us to flexibly calibrate any distribution shape by adjusting the number of mixtures as the data structure suggested. Two versions of SV-DPM jump models are proposed - static SV-DPM jump model and dynamic SV-DPM jump model where the dynamic DPM model incorporates the bipower variation in the means of distribution mixture. We show that our proposed model can provide better accuracy in detecting true jump arrivals in the simulation and can reveal the empirical distribution of jump sizes for financial indices - S&P 500 index (SPX), Gold and Silver index (XAU), Oil index (XOI) and Treasury bill (TYX). Our model performs especially better during volatile periods. More importantly, compared with parametric SVJ model, our model generates better predictive density of returns overall and at 10% and 90% quantiles. Significant gains can be observed after 2020 pandemic. From the option pricing and risk management perspective, we verify our proposed model performance with empirical S&P 500 IV smile and our model generated

IV fits better to the empirical IV of in-the-money (ITM) call options.

The paper is organized as follows. We specify our model structure and briefly introduce DPM in Section 2 and document Bayesian inference and sampling mechanism in Section 3. In Section 4, we investigate the model performance and robustness under different true data generating process (DGP). In particular, we compare the model performance with various true jump intensity and sizes. This gives us a summary about how the model performs corresponding to different market conditions. In Section 5, we verify the practical use of our model using empirical data. Specifically, we construct the model implied jump size distribution, predictive density of returns and the model generated IV smirk. We discuss the model drawbacks and future research in Section 6.

2 Modelling Jumps with Dirichlet Process Mixture

2.1 Stochastic Volatility and Jumps Process

Let p_t be the logarithm of the asset price P_t at time $t > 0$ and V_t be the latent variance. We assume a continuous time jump diffusion process,

$$\begin{aligned} dp_t &= \mu_t dt + (V_t)^{1/2} dW_t^p + dJ_t, \\ d(\log(V_t)) &= \kappa(\theta - \log(V_t))dt + \gamma dW_t^v \\ dJ_t &= Z_t dN_t \end{aligned} \tag{1}$$

where μ_t is the drift, (W_t^p, W_t^r) are standard Brownian motion and (κ, θ, γ) are parameters to describe the stochastic variance process $\log(V_t)$. $dJ_t = Z_t dN_t$ represents the price jump process with $P(dN_t = 1) = \delta dt$, $P(dN_t = 0) = (1 - \delta)dt$ and jump size Z_t. If the correlation ρ between dW_t^p and dW_t^ν is not zero, then there is leverage effect where the asset's volatility is negatively correlated with the asset's returns (Black, 1976; Christie, 1982). Previous literature has found the asymmetric feature of "leverage effects" and Yu (2005) documents the estimation of leverage effects in stochastic volatility models.

The specification in eq.(1) uses the logarithmic variance process, which automatically guarantees the positivity of V_t without imposing additional parameter restrictions. Moreover, the log volatility is typically assumed to follow an AR(1) Gaussian process, such as in Kim *et al.* (1998), which facilitates Bayesian inference and state space filtering. However, there is no close-form option pricing formula under this specification. By adding jumps, the model then governs both the mean-reverting variance process and the sudden extreme changes in price. The literature has shown that the specification of stochastic volatility with jumps

(SVJ) improves the model performance in option pricing (Bates, 1996, 2000; Eraker *et al.*, 2003; Eraker, 2004; Gatheral, 2011).

The assumption of jump size (Z_t) distribution is important, and conventionally, it is assumed to be one particular distribution only (see discussions in Merton (1976); Chan and Maheu (2002); Eraker *et al.* (2003); Eraker (2004); Maheu and McCurdy (2004); Kou and Wang (2004); Kou (2007)). Some more recent literature investigates the signed jumps where positive and negative jumps can be treated separately (Fulop *et al.*, 2015; Maneesoonthorn *et al.*, 2017). We propose a novel specification that models jump size distribution flexibly by using DPM, which brings the convenience of no pre-determined groups for constructing mixture. We relax the assumption of a single jump size distribution and allow the data to inform the number of groups in a mixture model. This model specification nests the parametric SVJ models and can accommodate the need of recovering empirical jump distribution. Our proposed model contributes to the existing literature on financial jump modeling and can bring benefits to the risk management and asset pricing.

2.2 Dirichlet Process Mixture for Jump Size

To address the rigidity of conventional assumptions on jump size, we propose to model jump size distribution using DPM proposed by Ferguson (1973). The jump size Z_t is assumed to follow

$$Z_t \sim \sum_{k=1}^{\infty} w_k F(\cdot), \tag{2}$$

where $F(\cdot)$ is the pdf function of a specified distribution and

$$\begin{aligned} w_k &= \nu_k \prod_{l=1}^{k-1}(1-\nu_l), \\ \nu_k &\sim \text{Beta}(1-\alpha). \end{aligned} \tag{3}$$

Sethuraman (1994) shows that the stick-breaking process (SBP) is a representation of the Dirichlet process and the concentration parameter α controls the distribution of weights across the infinite set of mixture components. A smaller value of α encourages fewer components and a larger α favors a larger number of components. Equation (3) can be simplified as $\boldsymbol{w} \sim \text{SBP}(\alpha)$.

In this paper, we specify the distribution $F(\cdot)$ as Gaussian and equation (2) becomes

$$Z_t \sim \sum_{k=1}^{\infty} w_k N(\mu_k, \sigma_k^2). \tag{4}$$

The mixture of normals is able to mimic the features of other distributions and for Bayesian inference purpose, normal distribution form conjugate priors, and using normal distributions can bring convenience for option pricing development. However, our proposed model structure does not necessarily require $F(\cdot)$ to be Gaussian, but it is preferred to keep the conjugacy for faster and easier Bayesian computation.

With this flexible jump size structure, the model nests the parametric SVJ model where jump size assumed to be normally distributed, but also allows the model to have as many mixture components as the data structure requires. In practice, the maximum number of groups can be managed to avoid redundancy and still keeps a reasonable level of flexibility. See Section 3 for the practical sampling scheme.

2.3 Discrete Time Static Model Representation

To perform Bayesian inference and in common with the literature (see Eraker *et al.* (2003); Fulop *et al.* (2015); Maneesoonthorn *et al.* (2017)), we draw inference in discrete time. Given the continuous-time model in eq.(1), we state a static SV-DPM jump model with the asset log return on day t being

$$r_t = p_{t+1} - p_t. \tag{5}$$

With jump size Z_t modeled as in equation (4), the full discrete-time model is defined as:
Measurement equation:

$$r_t = \exp(h_t/2)\epsilon_t + \Delta N_t Z_t, \tag{6}$$

State equations:

$$\begin{aligned} h_{t+1} &= \phi_0 + \phi_1 h_t + \sigma_h \eta_t, \\ \begin{pmatrix} \epsilon_t \\ \eta_t \end{pmatrix} &\sim N\left(0, \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}\right), \\ \Delta N_t &\sim \text{Bernoulli}(\delta) \\ Z_t &\sim \sum_{k=1}^{\infty} w_k N(\mu_k, \sigma_k^2), \\ \boldsymbol{w} &\sim \text{SBP}(\alpha), \end{aligned} \tag{7}$$

where for $t = 1, \ldots, T$, h_t is the latent log variance, $|\phi_1| < 1$, and ΔN_t represents the occurrence of price jump with $p(\Delta N_t = 1) = \delta$, and $p(\Delta N_t = 0) = 1 - \delta$.

2.4 Incorporating Volatility Measures in Dynamic Model

Supported by the empirical evidence, the sudden changes observed in financial returns can be driven by either the latent volatility process or price jumps, given volatility jumps excluded from the model. With the static DPM model, it is difficult to disintangle the connection among changes of return, volatility and price jumps (Duffie *et al.*, 2000; Barndorff-Nielsen and Shephard, 2002; Aıt-Sahalia, 2004; Eraker, 2004). To better identify the changes of volatility and price jumps, we incorporate the high frequency volatility measure in the jump size modeling. We propose to model the jump size mean proportional to high frequency volatility measures who indicates the movement of latent volatility.

The Bipower variation (BV_t) is used as it is well known that BV_t is a consistent estimator of integrated volatility (IV_t) and is defined as

$$BV_t = \frac{\pi}{2}(\frac{N}{N-1})\sum_{i=2}^{N}|r_{t_i}||r_{t_{i+1}}| \xrightarrow{p} IV_t, \text{ as } N \to \infty, \tag{8}$$

where $r_{t_i} = p_{t_{i+1}} - p_{t_i}$ is the ith of N observed returns between trading day t and $t+1$ (Barndorff-Nielsen and Shephard, 2004). However, since BV_t is constructed by high-frequency returns, it is often noisy due to the microstructure noise. Hence, we use the moving average of BV_t in the model.

The dynamic SV-DPM jump model is defined as following:

Measurement equation:

$$r_t = \exp(h_t/2)\epsilon_t + \Delta N_t Z_t, \tag{9}$$

State equations:

$$\begin{aligned} h_{t+1} &= \phi_0 + \phi_1 h_t + \sigma_h \eta_t, \\ \begin{pmatrix} \epsilon_t \\ \eta_t \end{pmatrix} &\sim N\left(0, \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}\right), \\ \Delta N_t &\sim \text{Bernoulli}(\delta) \\ Z_t &\sim \sum_{k=1}^{\infty} w_k N(\mu_k \bar{BV_t}^{1/2}, \sigma_k^2), \\ \boldsymbol{w} &\sim \text{SBP}(\alpha), \\ (\bar{BV_t})^{1/2} &= [\frac{1}{n}\sum_{m=1}^{n}(BV_{t-m})]^{1/2} \end{aligned} \tag{10}$$

Since extreme returns can arise from either price jumps or changes in volatility, the static DPM model faces challenges in adapting the jump-size distribution during periods of elevated volatility. In such episodes, both the observed returns and the latent volatility tend to be high, leaving only a small residual component to be attributed to the jump-diffusion term. Consequently, the model may incorrectly classify moderate return movements as jumps, even though they may not reflect genuinely abrupt or discontinuous price changes. Allowing jump size mean to be proportional to the latent volatility complicates the sampling schemes and with two latent processes entangled together, it is more difficult to identify either of them through the nonparametric techniques. With BV being a consistent estimator of the latent volatility, it proxies the volatility movements. Incorporating BV in the jump size distribution drives the jump size mean away from zero more acutely during high volatility days.

3 Bayesian Inference

For notion convenience, let $\Theta = \{\phi_0, \phi_1, \sigma_h^2, \rho\}$ and $\boldsymbol{\Phi} = \{\boldsymbol{w}, \boldsymbol{\mu}, \boldsymbol{\sigma^2}, \delta\}$ where $\boldsymbol{\mu} = \{\mu_1, \ldots, \mu_\infty\}$ and $\boldsymbol{\sigma^2} = \{\sigma_1^2, \ldots, \sigma_\infty^2\}$. Based on the model specified in Section 2.3, the joint posterior can be formulated in an expression as:

$$\begin{aligned} P(\boldsymbol{\Theta}, \boldsymbol{\Phi}, \boldsymbol{h}, \boldsymbol{\Delta N}, \boldsymbol{Z}|\boldsymbol{r}) \propto\ & P(\phi_0, \phi_1|\sigma_h^2)P(\sigma_h^2)P(\delta)P(\rho)P(h_0) \\ & \prod_{t=1}^{T}\Big[P(\Delta N_t|\delta)P(h_{t+1}, r_t|h_t, \phi_0, \phi_1, \sigma_h^2, \rho, \Delta N_t, Z_t) \\ & \prod_{k=1}^{\infty} P(\mu_k, \sigma_k^2)P(w_k)P(Z_t|\mu_k, \sigma_k^2)\Big] \end{aligned} \tag{11}$$

and the joint posterior of dynamic model described in Section 2.4 has the form of:

$$\begin{aligned} P(\boldsymbol{\Theta}, \boldsymbol{\Phi}, \boldsymbol{h}, \boldsymbol{\Delta N}, \boldsymbol{Z}|\boldsymbol{r}, (\boldsymbol{B\bar{V}})^{1/2}) \propto\ & P(\phi_0, \phi_1|\sigma_h^2)P(\sigma_h^2)P(\delta)P(\rho)P(h_0) \\ & \prod_{t=1}^{T}\Big[P(\Delta N_t|\delta)P(h_{t+1}, r_t|h_t, \phi_0, \phi_1, \sigma_h^2, \rho, \Delta N_t, Z_t) \\ & \prod_{k=1}^{\infty} P(\mu_k, \sigma_k^2)P(w_k)P(Z_t|\mu_k(B\bar{V}_t)^{1/2}, \sigma_k^2)\Big] \end{aligned} \tag{12}$$

Due to the complexity of state space models and the infinite mixture in the definition of DPM, proper sampling mechanisms need to be imposed.

3.1 Slice Sampler

A challenge of sampling from DPM is the infinite dimension. Various algorithms have been proposed to deal with this in practice. See discussions in MacEachern (1994); MacEachern and Müller (1998); Neal (2000); Ishwaran and James (2001, 2002); Walker (2007). As one of these algorithms, slice sampler has been widely used for DPM sampling, which uses an auxiliary variable to sample dominating yet finite number of components (Neal, 2003; Walker, 2007; Kalli *et al.*, 2011). Let u_t be the auxiliary variable and $u_t|\boldsymbol{w} \sim U(0, \boldsymbol{w})$. The joint density of Z_t and u_t is

$$\begin{aligned} P(Z_t, u_t|\cdot) &\propto \sum_{k=1}^{\infty} I(u_t < w_k) N(Z_t|\mu_k, \sigma_k^2), \\ &\propto \sum_{k \in A_t} N(Z_t|\mu_k, \sigma_k^2), \end{aligned} \tag{13}$$

where $A_t = \{k : u_t < w_k\}$. The conditional density of Z_t given u_t is

$$P(Z_t|u_t, \cdot) = \frac{1}{f(u_t)} \sum_{k \in A_t} N(Z_t|\mu_k, \sigma_k^2), \tag{14}$$

where $f(u_t) = \sum_{k=1}^{\infty} I(u_t < w_k)$ which is defined on $(0, \max(\boldsymbol{w}))$ (Walker, 2007). With slice sampler, we can "slice" the infinite components to dominating and finite K components and the posterior of $\boldsymbol{w}$ can be represented as $\boldsymbol{w} = \{w_1, w_2, \ldots, w_K, \bar{w_K}\} \sim \text{Dir}(n_1, n_2, \ldots, n_K, \alpha)$ where $n_k = \sum_{t=1}^{T} I(S_t^z = k)$, and S_t^z is the indicator of groups and α is the precision parameter for the residual weight $\bar{w_K}$ (Ferguson, 1973).

To reduce the extensive computation associated with state space models, we acknowledge the fact that the sampling of jump size Z_t is only needed when a jump is present ($\Delta N_t = 1$). Therefore, we allow $S_t^z = 0, 1, 2, \ldots, K$ and $P(S_t^z = 0) = w_0$ represents $\Delta N_t = 0$ where $w_0 \sim \text{Beta}(a, b)$ and $(1 - w_0) \sim \text{SBP}(\alpha)$. After integrating out ΔN_t, the joint posteriors are:

$$\begin{aligned} P(\boldsymbol{\Theta}, \boldsymbol{\Phi}, \boldsymbol{h}, \boldsymbol{S^z}, \boldsymbol{u}, \boldsymbol{Z}|\boldsymbol{r}) \propto\ & P(\phi_0, \phi_1|\sigma_h^2) P(\sigma_h^2) P(\boldsymbol{w}) P(h_0) \\ & \prod_{t=1}^{T} \Big[P(S_t^z|\boldsymbol{w}) P(u_t|S_t^z, \boldsymbol{w}) P(h_{t+1}, r_t|h_t, \phi_0, \phi_1, \sigma_h^2, \rho, \Delta N_t, Z_t) \\ & \prod_{S_t^z=1}^{K} P(\mu_{S_t^z}, \sigma_{S_t^z}^2) P(Z_t|\mu_{S_t^z}, \sigma_{S_t^z}^2) \Big] \end{aligned} \tag{15}$$

$$\begin{aligned} P(\boldsymbol{\Theta}, \boldsymbol{\Phi}, \boldsymbol{h}, \boldsymbol{S^z}, \boldsymbol{u}, \boldsymbol{Z}|\boldsymbol{r}, \boldsymbol{(B\bar{V})^{1/2}}) \propto\ & P(\phi_0, \phi_1|\sigma_h^2) P(\sigma_h^2) P(\boldsymbol{w}) P(h_0) \\ & \prod_{t=1}^{T} \Big[P(S_t^z|\boldsymbol{w}) P(u_t|S_t^z, \boldsymbol{w}) P(h_{t+1}, r_t|h_t, \phi_0, \phi_1, \sigma_h^2, \rho, \Delta N_t, Z_t) \\ & \prod_{S_t^z=1}^{K} P(\mu_{S_t^z}, \sigma_{S_t^z}^2) P(Z_t|\mu_{S_t^z} (B\bar{V}_t)^{1/2}, \sigma_{S_t^z}^2) \Big] \end{aligned} \tag{16}$$

3.2 Sampling Latent Volatility in Presence of Leverage Effect

Incorporating the leverage effect in the SV model complicates the sampling of the latent volatility, since the correlation between return and volatility innovations is introduced. As the conditional distribution of the latent volatility given the data is no longer Gaussian and does not remain independent from the return innovation, the sampling scheme with seven component mixtures in Kim *et al.* (1998) is no longer applied. Therefore, to properly sample the latent volatility with leverage effect imposed, the random blocking Metropolis-Hastings (MH) algorithm is adopted for sampling latent volatility.

The proposed density is the seven component mixture of Gaussian distributions proposed by Kim *et al.* (1998) with no leverage effect. With the leverage effect imposed in the model as eq.(9) - (10) and by the rule of changing variables, we have the target density as

$$\begin{aligned} P(h_{t+1}|h_t, y_t, \Theta, \cdot)P(y_t|h_t, \Theta, \cdot) &= \frac{P(\eta_t|\epsilon_t)P(\epsilon_t)}{|e^{h_t/2}||\sigma_h|}, \text{ where} \\ y_t &= r_t - \Delta N_t Z_t \end{aligned} \tag{17}$$

With $\epsilon_t^c = y_t/e^{h_t^c/2}$ and $\eta_t^c = (h_{t+1}^c - \phi_0 - \phi_1 h_t^c)/\sigma_h$, the MH ratio is then defined as:

$$\begin{aligned} MHR &= \min\left\{1, \prod_{t=\text{start}}^{\text{end}} \frac{P^a(y_t|h_t)P^a(h_{t+1}|h_t)}{P^a(y_t|h_t^c)P^a(h_{t+1}^c|h_t^c)} \cdot \frac{P(y_t|h_t^c)P(h_{t+1}^c|h_t^c, y_t)}{P(y_t|h_t)P(h_{t+1}|h_t, y_t)}\right\} \\ &= \min\left\{1, \prod_{t=\text{start}}^{\text{end}} \frac{P^a(y_t|h_t)P^a(h_{t+1}|h_t)}{P^a(y_t|h_t^c)P^a(h_{t+1}^c|h_t^c)} \cdot \frac{P(\epsilon_t^c)P(\eta_t^c|\epsilon_t^c)}{P(\epsilon_t)P(\eta_t|\epsilon_t)} \cdot \frac{|e^{-h_t^c/2}|}{|e^{-h_t/2}|}\right\} \end{aligned} \tag{18}$$

where the $P^a(\cdot)$ represents the proposed density from the seven component mixture by Kim *et al.* (1998) and the $P(\cdot)$ represents the target density derived from eq.(17).

For sampling the leverage effect parameter ρ, from the conditional distribution $\eta_t|\epsilon_t \sim N(\rho\epsilon_t, 1-\rho^2)$, a linear regression can be derived as in the following equation:

$$\begin{aligned} \eta_t &= \rho\epsilon_t + \sqrt{1-\rho^2} z_t, \text{ where } z_t \sim N(0,1), \\ \frac{h_{t+1} - \phi_0 - \phi_1 h_t}{\sigma_h} &= \rho\epsilon_t + (1-\rho^2)^{1/2} z_t, \\ h_{t+1} - \phi_0 - \phi_1 h_t &= \sigma_h \rho \epsilon_t + \sigma_h (1-\rho^2)^{1/2} z_t, \\ \tilde{h}_{t+1} &= \beta_\rho \epsilon_t + \sigma_\rho z_t \end{aligned} \tag{19}$$

The β_z, σ_z can be drawn as a Normal-Inverse Gamma conjugate prior and parameter ρ can be solved from it.

The full Gibbs sampling algorithm is as following with total M iterations:

Algorithm 1 Full Gibbs sampling algorithm

Require: prior, $\boldsymbol{r}$

Initialize $(i=0)$ $\boldsymbol{\Theta}, \boldsymbol{S}$ and $\boldsymbol{h}$; $\triangleright$ $\boldsymbol{S}$ is the indicator for seven component mixture (Kim *et al.*, 1998).

Initialize $(i=0)$ $\boldsymbol{(\mu_{S^z_t}, \sigma^2_{S^z_t}), S_z}$ and $\boldsymbol{Z}$.

For each MCMC iteration i

1. Sample $\boldsymbol{w^{(i)}|S_z^{(i-1)}}$
2. Sample $\boldsymbol{u^{(i)}|w^{(i)}, S_z^{(i-1)}}$
3. Sample $\boldsymbol{(\mu_{S^z_t}, \sigma^2_{S^z_t})^{(i)}|S_z^{(i-1)}, Z^{(i-1)}}$
4. Sample $\boldsymbol{S_z^{(i)}|Z^{(i-1)}, (\mu_{S^z_t}, \sigma^2_{S^z_t})^{(i)}, w^{(i)}, u^{(i)}}$
5. Sample $\boldsymbol{Z^{(i)}|h^{(i-1)}, r, S_z^{(i)}, (\mu_{S^z_t}, \sigma^2_{S^z_t})^{(i)}}$
6. Sample $\boldsymbol{h^{(i)}|r, Z^{(i)}, S^{(i-1)}, \Theta^{(i-1)}}$
7. Sample $\boldsymbol{\Theta^{(i)}|h^{(i)}}$
8. Sample $\boldsymbol{S^{(i)}|h^{(i)}, r}$
9. Go to step (1) and repeat for M iterations.

4 Simulation Analysis

4.1 Simulation Design

We define the true data generating process (DGP) as following and generate $t = 1, 2, \dots, T$ observations from:

$$
\begin{aligned}
r_t &= \exp(h_t/2)\epsilon_t + \Delta N_t Z_t, \\
h_t &= -0.06 + 0.94 h_{t-1} + 0.38\eta_t \\
\begin{pmatrix} \epsilon_t \\ \eta_t \end{pmatrix} &\sim N\left(0, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right), \\
\log(BV_t) &= h_t + 0.4\epsilon_t^{BV} \\
\bar{BV}_t &= \frac{1}{5}\sum_{m=1}^{5} BV_{t-m} \\
\Delta N_t &\sim \text{Bernoulli}(\delta), \\
Z_t &\sim 0.6 \times N(-\mu_z e^{h_t/2}, 0.8) + 0.4 \times N(\mu_z e^{h_t/2}, 0.8),
\end{aligned}
\tag{20}
$$

where $\epsilon_t^r, \epsilon_t^h$ follow standard normal distributions and $T = 5000$. Acknowledging the fact that jumps being extreme changes and the literature findings from Maneesoonthorn *et al.*

(2017), we set the true jump size Z_t to be a mixture of normal distributions to mimic the bimodal distribution. With the variance of each component being relatively small, we avoid generating jump sizes around 0. A slightly heavier weight is imposed on the negative component for Z_t to mimic the negatively skewed return distribution observed empirically.

We compare the performance among parametric SVJ model, static SV-DPM jump model and dynamic SV-DPM jump model. The specification of each model is defined as in the following table. We are interested in the model performance and its robustness under different

Table 1: Model specifications of parametric SVJ model ($\mathcal{M}_1$), static SV-DPM jump model ($\mathcal{M}_2$) and dynamic SV-DPM jump model ($\mathcal{M}_3$).

Models	**Specification**
$\mathcal{M}_1$: Parametric SVJ model	$\begin{aligned} r_t &= \exp(h_t/2)\epsilon_t + \Delta N_t Z_t, \\ h_t &= \phi_0 + \phi_1 h_{t-1} + \sigma_h \eta_t, \\ \Delta N_t &\sim \text{Bernoulli}(\delta), \\ Z_t &\sim N(\mu_z, \sigma_z^2). \end{aligned}$
$\mathcal{M}_2$: Static SV-DPM jump model	$\begin{aligned} r_t &= \exp(h_t/2)\epsilon_t + \Delta N_t Z_t, \\ h_t &= \phi_0 + \phi_1 h_{t-1} + \sigma_h \eta_t, \\ \Delta N_t &\sim \text{Bernoulli}(\delta), \\ Z_t &\sim \sum_{k=1}^{\infty} w_k N(\mu_k, \sigma_k^2). \end{aligned}$
$\mathcal{M}_3$: Dynamic SV-DPM jump model	$\begin{aligned} r_t &= \exp(h_t/2)\epsilon_t + \Delta N_t Z_t, \\ h_t &= \phi_0 + \phi_1 h_{t-1} + \sigma_h \eta_t, \\ \Delta N_t &\sim \text{Bernoulli}(\delta), \\ Z_t &\sim \sum_{k=1}^{\infty} w_k N(\mu_k \bar{BV}_t^{1/2}, \sigma_k^2), \\ \bar{BV}_t &= \frac{1}{5}\sum_{m=1}^{5} BV_{t-m}. \end{aligned}$

market conditions. Therefore, we allow the true DGP jump intensity to vary from $\delta = 1\%$ to $\delta = 15\%$, corresponding to low and high volatile time periods in the financial market. In addition, we acknowledge the fact that there are small jumps like macroeconomic news, and large jumps, such as 2008 Global financial crisis (GFC) and 2020 pandemic. By allowing the jump size mean $\pm\mu_z e^{h_t/2}$ to change from $\pm 2e^{h_t/2}$ to $\pm 3.5e^{h_t/2}$, we mimic the jump size changes in the financial market. Specifically, we set the range of true jump intensity as $\delta = \{1\%, 5\%, 10\%, 15\%\}$ and the true jump size mean as $\pm\{2e^{h_t/2}, 2.5e^{h_t/2}, 3e^{h_t/2}, 3.5e^{h_t/2}\}$. For simulation analysis, we deliberately choose non-informative priors for all parameters.

The prior specification can be found in Table 9 of the Appendix.

4.2 Simulation Results

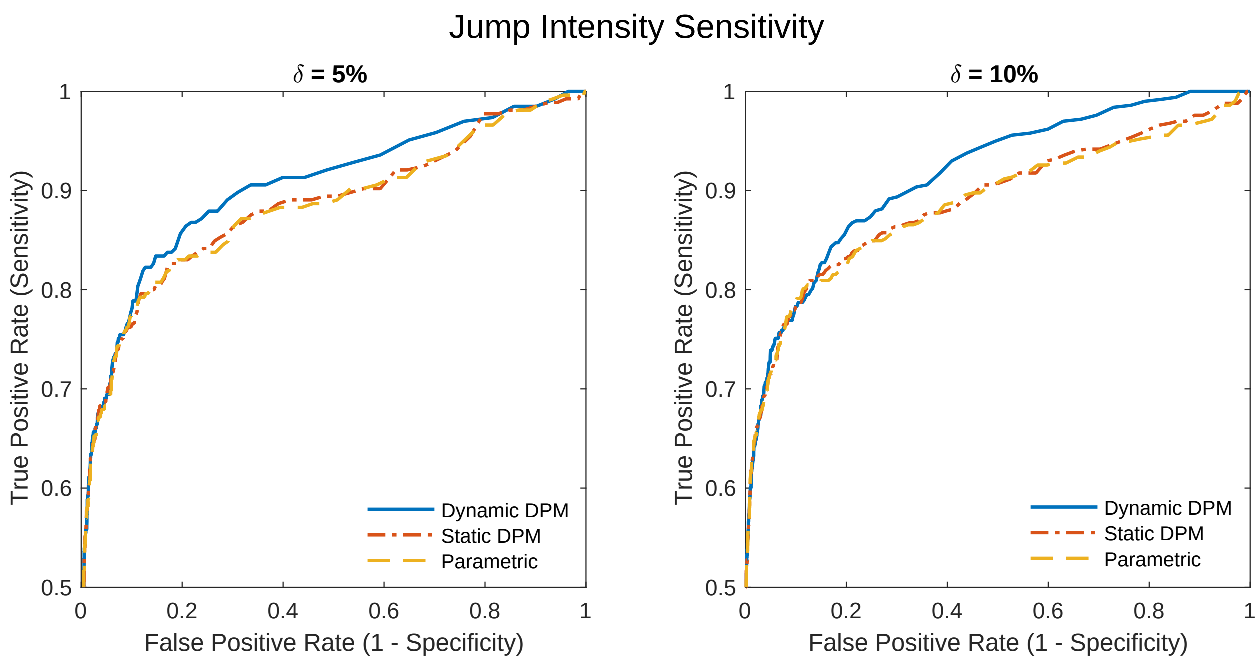
One of our main focuses is the estimation of jump occurrence. Therefore, we compare the model performance in jump occurrence estimation using the proportion of a model identified jump given there is a true jump (true positive rate), and the proportion of a model identified jump given there is no true jump (false positive rate). In addition, we compare the marginal posterior of jumps with the true jump density. To reduce the number of cases we present, we choose to elaborate on two scenarios. First, the jump intensity δ is fixed at 10% while the mean of jump size changes. Second, the true jump size mean is fixed at $\pm 3e^{h_t/2}$ while δ changes.

Table 2: Estimated parameter values of parametric SVJ model ($\mathcal{M}_1$), static SV-DPM jump model ($\mathcal{M}_2$) and the dynamic SV-DPM jump model ($\mathcal{M}_3$). The posterior mean and 90% posterior interval are reported in the table.

Simulation Estimation Results				
Parameters	**True Value**	**Parametric SVJ model**	**Static SV-DPM jump model**	**Dynamic SV-DPM jump model**
ϕ_0	-0.06	-0.0396	-0.0394	-0.039
		(-0.0488, -0.0305)	(-0.0491, -0.0300)	(-0.0494, -0.0295)
ϕ_1	0.94	0.9621	0.962	0.9625
		(0.9553, 0.9694)	(0.9550, 0.9694)	(0.9552, 0.9695)
σ_h	0.38	0.2758	0.2762	0.2721
		(0.2634, 0.2892)	(0.2631, 0.2899)	(0.2584, 0.2842)
δ	0.1	0.1236	0.1236	0.097
		(0.1093, 0.1384)	(0.1101, 0.1385)	(0.0857, 0.1096)
K	2	-	1	2
			[1,7]	[2,10]
μ_z	-3, 3	-0.2423	-0.2244	-3.0326, 3.0409
		(-0.3574, -0.1250)	(-0.4128, -0.0453)	(-3.3119, -2.7177); (2.6363, 3.4363)
σ_z^2	0.8944, 0.8944	2.1243	2.1254	0.8527, 0.9708
		(1.9739, 2.2956)	(1.9641, 2.2932)	(0.6925, 1.0199); (0.7859, 1.1873)

Table 2 reports the estimated parameter values for moderate volatile market set up where true jump intensity $\delta = 10\%$ and the true jump size mean are $\pm 3e^{h_t/2}$. Parametric SVJ model and static DPM model fail to identify the two components of the mixture stated in the true DGP and obtain very similar parameter estimations in jumps. Due to the model specification, parametric SVJ model lacks of flexibility and results in an jump size estimation with slightly negative mean and large variance to accommodate both positive and negative jumps. Static DPM model in structure can estimate multiple components and has the number of groups K ranging from 1 to 7, it struggles to distinguish the jump variation from the volatility variation when the true jump size is proportional to volatility and result in identifying only one group for most of the time. Meanwhile, dynamic DPM model is less misspecified and can recover the true parameters of jumps with extreme jump size mean and

Figure 1: Receiver Operating Characteristics (ROC) curves for jump intensity analysis. The jump intensity of the true DGPs is 5% (left panel) and 10% (right panel) and the true $\mu_z = 3$. The yellow dashed line is the ROC curve of parametric jump model ($\mathcal{M}_1$), the red dot-dashed line represents the ROC curve of static SV-DPM jump model ($\mathcal{M}_2$) and the blue solid line represents the ROC curve of dynamic SV-DPM jump model ($\mathcal{M}_3$).



small variance, even when $\bar{BV}_t$ is used to approximate h_t.

4.2.1 Jump Occurrence

We investigate further about the robustness of dynamic DPM model with various true DGPs as mentioned before. Figure 1 represents the Receiver Operating Characteristics (ROC) curve. With the true positive rate (sensitivity) plotted against the false positive rate (1−specificity), the closer the curve to the upper left corner of the graph, the higher accuracy of the test. Note that we do not present an asymptotic hypothesis test here, but the ROC curves effectively shows how accuracy each model is in identifying true jumps. The true jump size mean is fixed at $\pm 3e^{h_t/2}$ while each panel of Figure 1 represents different true jump intensity δ. When the true jump size being three standard deviation away from its mean, all three models achieve high sensitivity for a wide range of false-positive rates, suggesting robust ability to correctly identify true jumps. Comparing the right panel with the left one, as a higher proportion of jumps presented in the true DGP, it is easier for all three models to detect jumps more correctly. The static DPM model provides almost identical accuracy as parametric model in detecting true jumps, when $\delta = 5\%$. As the

Table 3: Area under the ROC curve (AUC) for both jump intensity analysis and jump size analysis under simulation. The larger AUC, the more accurate the model is in terms of detecting true jumps. The bold numbers are the maximum of each cases.

Area under the ROC curve (AUC)			
	Dynamic DPM	Static DPM	Parametric SVJ
A: Jump Intensity Analysis ($\mu_z = 3$)			
$\delta = 1\%$	0.8960	0.8791	**0.9043**
$\delta = 5\%$	**0.9054**	0.8883	0.8819
$\delta = 10\%$	**0.9004**	0.8816	0.8788
$\delta = 15\%$	**0.9208**	0.9018	0.9018
B: Jump Size Analysis ($\delta = 10\%$)			
$\mu_z = 2$	0.7966	0.7932	**0.7974**
$\mu_z = 2.5$	**0.8491**	0.8270	0.8287
$\mu_z = 3$	**0.9004**	0.8816	0.8788
$\mu_z = 3.5$	**0.9363**	0.9160	0.9171

true jump intensity increases to 10%, static DPM model is slightly better than parametric model. Dynamic DPM outperforms the other two models for both cases and as the true jump intensity increases, the improvement becomes more obvious.

Table 3 reports the area under ROC curve (AUC). The larger the value is, the more accurate the model is in terms of detecting true jumps. The bolded numbers are the maximum AUC among three models for each true DGP case. Panel A of Table 3 corresponds to jump intensity analysis plotted in Figure 1. Even though parametric SVJ model has the largest AUC when true $\delta = 1\%$, the difference between parametric model and dynamic DPM model is very small. Under low volatile market condition, it is slightly less favourable for using DPM models due to small sample problem. As the true jump intensity increases, dynamic DPM model achieves the largest AUC. Panel B documents the AUC when the true jump intensity is fixed at $\delta = 10\%$ and μ_z changes. When $\mu_z = 2$, it is hard for all models to distinguish the variation of stochastic volatility and jumps, resulting an AUC being around 0.8. Parametric SVJ model performs the best but the difference is very slight. However, similar to the jump intensity sensitivity analysis, it is obvious that dynamic DPM outperforms the other two models as the true jump size moves away from zero. The improvement from using dynamic DPM becomes more obvious as μ_z gets larger. Static DPM model shows similar AUC level to the parametric model.

5 Empirical Analysis

5.1 Data Overview

The previous simulation analysis shows the superiority of dynamic DPM model over parametric SVJ and static DPM. We thereby investigate the practical use of dynamic DPM model using empirical data. Four financial indices from 3 January, 2000 to 31 December, 2021 are used in this section: S&P500 (SPX), Treasury Yield 30 years bond (TYX), Gold Index (XAU) and Amex Oil Index (XOI). The daily log returns are aggregated from 5-min high-frequency data of each index. Table 4 summarizes the descriptive statistics of indices returns and the jump hypothesis test proposed by Barndorff-Nielsen and Shephard (2006) (BNS test). The test statistics is defined as:

$$BNS_t = \frac{(RV_t - BV_t)/RV_t}{\sqrt{(\pi^2/4 + \pi - 5)N^{-1}\max(1, TP_t/BV_t^2)}}, \tag{21}$$

where

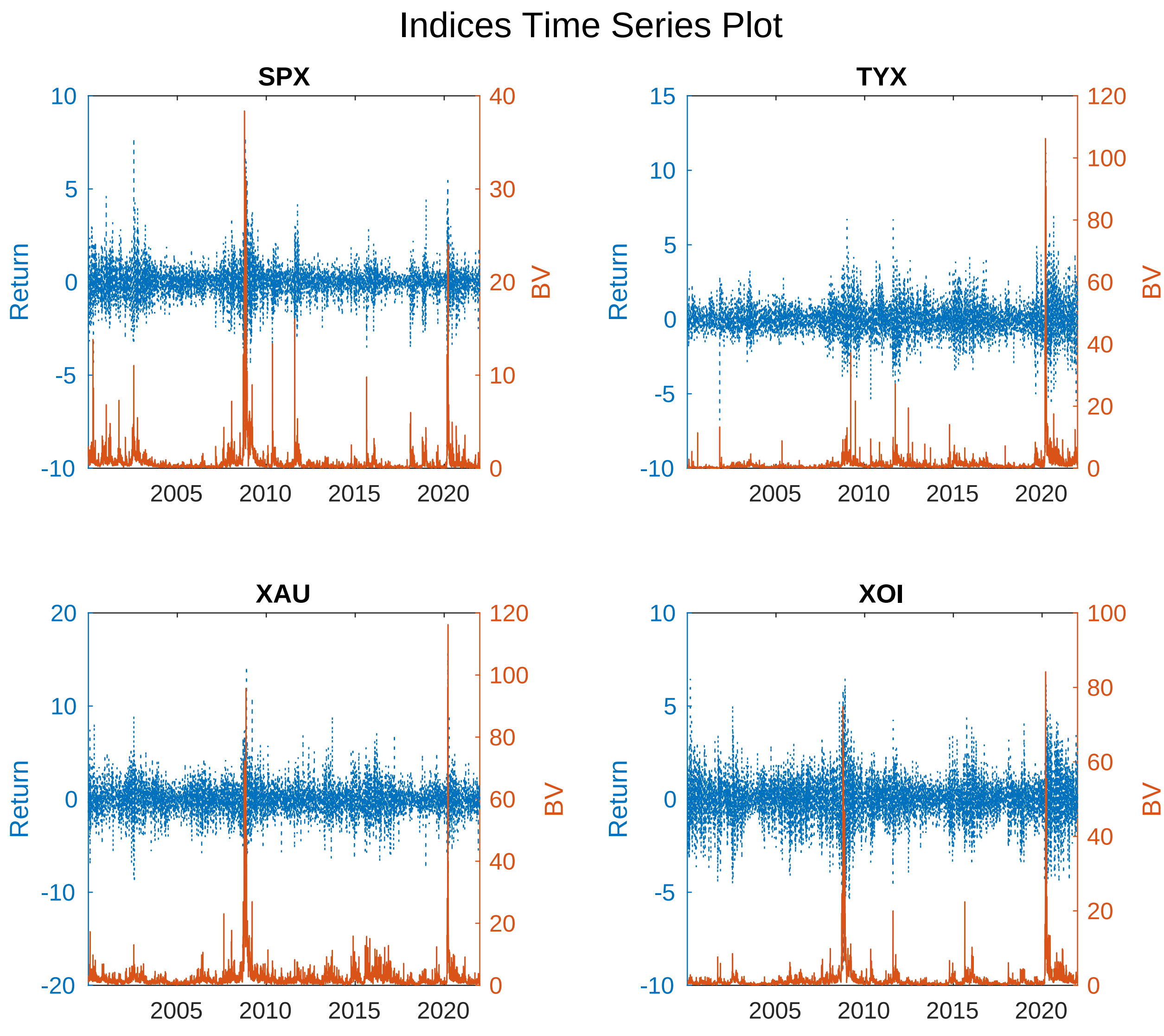
$$\begin{aligned} TP_t &= [2^{2/3}\Gamma(7/6)\Gamma(1/2)^{-1}]^{-3}\left(\frac{N^2}{N-2}\right)\sum_{i=3}^{N}|r_{t_{i-2}}|^{4/3}|r_{t_{i-1}}|^{4/3}|r_{t_i}|^{4/3}, \\ RV_t &= \sum_{i=1}^{N}|r_{t_i}|^2. \end{aligned} \tag{22}$$

Table 4: Descriptive statistics (mean, standard deviation, skewness and kurtosis) of indices log returns.

Descriptive Statistics of Indices					
	Mean (%)	Std. (%)	Skewness	Kurtosis	BNS test (%)
SPX	0.6200	90.1000	0.0551	12.1733	7.4100
TYX	-0.2100	116.9900	1.1500	19.1961	14.2700
XAU	-5.3300	179.4800	0.0359	7.6912	4.3800
XOI	-1.7300	128.4000	-1.2556	22.3741	6.7600

The mean of four indices are all around zero and they all have large kurtosis, which aligns with financial return's stylized fact about heavy tails. TYX and XOI are relatively more skewed than SPX and XAU. Based on the results from BNS test, SPX, XAU and XOI have roughly 4% to 7% jumps, while the jump intensity of TYX is relatively higher - 14%. Figure 2 shows how each index and its BV changes over time via time series plots. The blue dot line represents the historical returns of each index and the red solid line is the associated

Figure 2: Time series plots of returns and BV for four indices - SPX, TYX, XAU and XOI. The blue dotted line represents the daily returns and the red solid line are BV computed using 5-min high frequency data.



BV. From this plot, we can see that the extreme returns can be driven by large volatility, measured by large BV, instead of price jumps. This further confirms our motivation of proposing dynamic DPM model.

5.2 Estimation Results

We use the data from 3 January, 2000 to 2 May, 2014 for in-sample estimation under this section. The estimation results for dynamic DPM and dynamic DPM with leverage effects are both provided for comparison. The estimated parameters for dynamic DPM is documented in Table 5. This table shows the estimated parameters for each index, the mode of the number of components K and the associated mean and variance of each component.

From Table 5, for all indices, dynamic DPM model is able to identify two well-separated components with one positive and one negative jump size mean, and the variance of each jump component is relatively small. Although the number of mixture components is infinite in theory, in practice it is unnecessary to allow an unbounded number. For computational efficiency, the maximum of the components K is set to be 10. Over MCMC iterations, K changes and can range from 2 to 10. For SPX, TYX and XAU, K does hit its maximum value. In addition, the posterior mean of δ serves as an estimation of jump intensity. The estimated jump intensity for all indices are similar, with SPX has the lowest intensity at around 11% and XAU has the highest jump intensity around 13%, which is reasonable based on empirical findings in the literature.

Table 6 shows the estimated parameters for dynamic DPM model with leverage effects. All indices show a negative posterior mean of correlation $\hat{\rho}$. We can see that the estimated parameters do not show a big difference to the ones shown in Table 5. The model can still identify two separate groups for the upper and lower tails of the returns. However, as both leverage effect and the jump contribute to the asymmetric feature of the return distribution, the estimated σ_z becomes slightly larger.

5.3 Predictive Density

To demonstrate the practical usefulness of the proposed DPM models, we evaluate its out-of-sample performance by computing one-step-ahead predictive densities using an expanding-window scheme. Specifically, the initial estimation window spans 3 January 2000 to 2 May 2014, after which the sample is expanded sequentially by one observation at each step to generate the next one-step-ahead forecast. This procedure continues until the end of the sample, yielding a total of roughly 1700 evaluation periods, with the exact number varying across indices. For each MCMC iteration i, we obtain sample draws for parameters and

Table 5: Estimated parameters and jump components K, and the corresponding mean and variance of each component. We report the posterior mean and 90% posterior interval or mode and range for K.

Estimation Results		
Parameters	Posterior Mean/Mode (for K)	90% Posterior Interval/Range (for K)
	SPX	
ϕ_0	-0.0129	(-0.0198, -0.0067)
ϕ_1	0.9862	(0.9808,0.9915)
σ_h	0.1568	(0.1440, 0.1712)
δ	0.1071	(0.0757,0.1435)
K	2	[2,10]
μ_z	-1.5656; 1.2220	(-1.9709, -1.1900); (0.9236, 1.5885)
σ_z	0.4787; 0.4022	(0.3703, 0.5934); (0.3192, 0.5044)
	TYX	
ϕ_0	-0.0057	(-0.0100, -0.0016)
ϕ_1	0.9885	(0.9835, 0.9931)
σ_h	0.1196	(0.1093, 0.1301)
δ	0.1107	(0.0744, 0.1538)
K	2	[2,10]
μ_z	-1.2943; 1.5057	(-1.8216,-0.8522); (1.2094, 1.8179)
σ_z	0.5380; 0.4849	(0.3946, 0.7217); (0.3668, 0.6322)
	XAU	
ϕ_0	0.0072	(0.0029, 0.0115)
ϕ_1	0.9836	(0.9773, 0.9892)
σ_h	0.1239	(0.1133, 0.1359)
δ	0.1341	(0.1031, 0.1653)
K	2	[2, 10]
μ_z	-1.6369; 2.0420	(-1.9333, -1.3232); (1.6303, 2.4210)
σ_z	0.6294; 0.5586	(0.4344, 0.8770); (0.4038, 0.7638)
	XOI	
ϕ_0	-0.0034	(-0.0073, 0.0006)
ϕ_1	0.9857	(0.9804, 0.9905)
σ_h	0.1328	(0.1219, 0.1443)
δ	0.1249	(0.0876, 0.1645)
K	2	[2,6]
μ_z	-1.7014; 1.1320	(-2.0487, -1.3210); (0.8083, 1.4869)
σ_z	0.5556; 0.5728	(0.4019, 0.7758); (0.4238, 0.7506)

Table 6: Estimated parameters and jump components K, and the corresponding mean and variance of each component. We report the posterior mean and 90% posterior interval or mode and range for K.

Estimation Results with Leverage Effects		
Parameters	Posterior Mean/Mode (for K)	90% Posterior Interval/Range (for K)
	SPX	
ϕ_0	-0.0154	(-0.0226, -0.0086)
ϕ_1	0.9825	(0.9767,0.9880)
σ_h	0.1712	(0.1572, 0.1857)
δ	0.1033	(0.0724,0.1368)
ρ	-0.1374	(-0.1691, -0.1045)
K	2	[2,10]
μ_z	-1.5253; 1.2407	(-1.9124, -1.1445); (0.9283, 1.5846)
σ_z	0.4832; 0.4068	(0.3710, 0.6078); (0.3262, 0.5113)
	TYX	
ϕ_0	-0.0074	(-0.0122, -0.0028)
ϕ_1	0.9864	(0.9812, 0.9915)
σ_h	0.1311	(0.1193, 0.1428)
δ	0.1185	(0.0806, 0.1596)
ρ	-0.0664	(-0.1068, -0.0250)
K	2	[2,7]
μ_z	-0.8012; 0.9850	(-1.6998,1.5821); (-1.3972, 1.7375)
σ_z	0.4993; 0.4971	(0.3720, 0.6598); (0.3803, 0.6587)
	XAU	
ϕ_0	0.0080	(0.0033, 0.0129)
ϕ_1	0.9815	(0.9748, 0.9877)
σ_h	0.1321	(0.1202, 0.1459)
δ	0.1341	(0.1040, 0.1691)
ρ	-0.0427	(-0.0850, -0.0008)
K	2	[2, 10]
μ_z	-1.6188; 2.0436	(-1.9136, -1.3015); (1.6742, 2.3848)
σ_z	0.6411; 0.5557	(0.4353, 0.9273); (0.3991, 0.7628)
	XOI	
ϕ_0	-0.0041	(-0.0084, 0.0000)
ϕ_1	0.9820	(0.9764, 0.9875)
σ_h	0.1454	(0.1332, 0.1588)
δ	0.1233	(0.0893, 0.1608)
ρ	-0.0979	(-0.1348, -0.0593)
K	2	[2,7]
μ_z	-1.7012; 1.1223	(-2.0699, -1.3195); (0.7904, 1.4765)
σ_z	0.5600; 0.5735	(0.3978, 0.7604); (0.4318, 0.7552)

states and let all the information known at time t be $\mathcal{F}_t$ and construct the one-step ahead predictive density as:

$$\begin{aligned} \text{Without leverage: } \hat{p}^{(i)}(h_{t+1}|\mathcal{F}_t) &= N\Big(\phi_0^{(i)} + \phi_1^{(i)} h_t^{(i)}, (\sigma_h^2)^{(i)}\Big) \\ \text{With leverage: } \hat{p}^{(i)}(h_{t+1}|\mathcal{F}_t) &= N\Big(\phi_0^{(i)} + \phi_1^{(i)} h_t^{(i)} + (\sigma_h \rho \epsilon_t)^{(i)}, (\sigma_h^2(1-\rho^2))^{(i)}\Big) \end{aligned} \tag{23}$$

$$\begin{aligned} \hat{p}^{(i)}(r_{t+1}|\mathcal{F}_t, (\bar{BV}_{t+1}^{(i)})^{1/2}, h_{t+1}) = (1-\delta^{(i)}) N(0, e^{h_{t+1}^{(i)}}) + \\ \delta^{(i)} \sum_{k=1}^{K^{(i)}} w_k^{(i)} N\Big(\mu_k^{(i)} \times (\bar{BV}_{t+1}^{(i)})^{1/2}, e^{h_{t+1}^{(i)}} + (\sigma_k^2)^{(i)}\Big), \end{aligned} \tag{24}$$

where

$$\begin{aligned} \hat{\log}(BV_{t+1}^{(i)}) &= \hat{h}_{t+1}^{(i)} = \phi_0^{(i)} + \phi_1^{(i)} h_t^{(i)} + \sigma_h^{(i)} \epsilon_t^h \\ (\bar{BV}_{t+1}^{(i)})^{1/2} &= \frac{1}{5}[\sum_{g=1}^{4} BV_{t+1-g} + \hat{BV}_{t+1}^{(i)}]. \end{aligned} \tag{25}$$

The predictive density of return is obtained as:

$$\hat{p}(r_{t+1}|\mathcal{F}_t) = \frac{1}{M} \sum_{i=1}^{M} \hat{p}^{(i)}(r_{t+1}|\mathcal{F}_t, (\bar{BV}_{t+1}^{(i)})^{1/2}, h_{t+1}), \tag{26}$$

where M is the total number of MCMC iterations.

The predictive density is evaluated by strictly proper scoring rules (Gneiting and Raftery, 2007). The log score (LS) and the censored logarithmic score (CLS) are computed so that we evaluate the predictive ability of our model for the whole density and for the tails of return distribution (Giacomini and Komunjer, 2005; Geweke and Amisano, 2010; Diks *et al.*, 2011; Gneiting and Ranjan, 2011). Let the realization of future observation be r_{t+1}^o, and the density forecast is P_{t+1}. The scoring rules implemented are defined as following:

$$S_{LS}(P_{t+1}, r_{t+1}^o) = \log(p(r_{t+1}^o|\mathcal{F}_t)) \tag{27}$$

$$S_{CLS}(P_{t+1}, r_{t+1}^o) = I(r_{t+1}^o \in A) \log\left(p(r_{t+1}^o|\mathcal{F}_t)\right) + I(r_{t+1}^o \in A^c) \log\left(\int_{r \in A^c} p(r|\mathcal{F}_t) dr\right), \tag{28}$$

where A is the region of interest and A^c is its complement. To illustrate the predictive performance of DPM models, compared with the parametric SVJ model, the score difference is used (Geweke and Amisano, 2010). Let P_{t+1}^m be the predictive density obtained from either the dynamic or static DPM models, and P^b_{t+1} be the benchmark parametric SVJ model.

$$\begin{aligned} D^{LS}_{t+1} &= S_{LS}(P^m_{t+1}, r^o_{t+1}) - S_{LS}(P^b_{t+1}, r^o_{t+1}) \\ D^{CLS}_{t+1} &= S_{CLS}(P^m_{t+1}, r^o_{t+1}) - S_{CLS}(P^b_{t+1}, r^o_{t+1}). \end{aligned} \tag{29}$$

Figure 3 shows the change of cumulative score difference between DPM models and parametric SVJ model when the leverage effect is not considered. The blue solid line is the cumulative score difference between dynamic DPM and parametric SVJ model and the red dashed line shows the cumulative difference between static DPM and parametric SVJ model. LS, CLS of 10% and 90% quantile are documented for four financial indices. If the difference is larger than 0, then it shows that the predictive density from DPM models outperforms parametric SVJ model's. For SPX, XAU and XOI, the cumulative score difference between dynamic DPM and parametric model is almost always above zero and the difference increases over time. It suggests that the dynamic DPM can provide better predictions for the whole return distribution and for the 10% and 90% quantile of the distribution. Notably, there is a big improvement during 2020 pandemic, especially in predicting the 10% and 90% quantile. For XOI, dynamic DPM model provides better predictions of return distribution and of 10% quantile, but for 90% quantile, the difference is small and after 2019, parametric model is constantly better than dynamic DPM. Similar to XOI, there is no constant outperformance of dynamic DPM for TYX. However, there is a significant improvement after 2020, for LS, CLS at 10% and 90%. On the other hand, static DPM model gives unstable and mixed performance. For XAU, it outperforms parametric model for predicting the whole distribution and 90% quantile but for predicting 10% quantile, the performance of static DPM and parametric SVJ is very similar. For XOI, static DPM is constantly worse than parametric SVJ for all three scoring rules. In addition, for SPX and TYX, static DPM performs better in predicting 90% quantile but gives similar predictions for 10% quantile and the whole distribution.

Figure 4 shows the change of cumulative score difference between DPM models and parametric SVJ model when the leverage effect is incorporated in the model. The blue solid line is the cumulative score difference between dynamic DPM and parametric SVJ model and the red dashed line shows the cumulative difference between static DPM and parametric SVJ model. After considering leverage effect, dynamic DPM model still obtains better performance than the parametric SVJ model. For SPX and XOI, dynamic DPM performs much better than parametric model in predicting the lower 10% quantile, but does not produce consistently better predictions at the 90% quantile. However, the rising in the cumulative difference of predictive scores after 2020 pandemic shows that we obtain

Figure 3: Cumulative score difference between DPM models and parametric SVJ model when the leverage effect is not considered. The blue solid line is the cumulative score difference between dynamic DPM and parametric SVJ model and the red dashed line shows the cumulative score difference between static DPM and parametric SVJ model. LS, CLS of 10% and 90% quantile are documented.

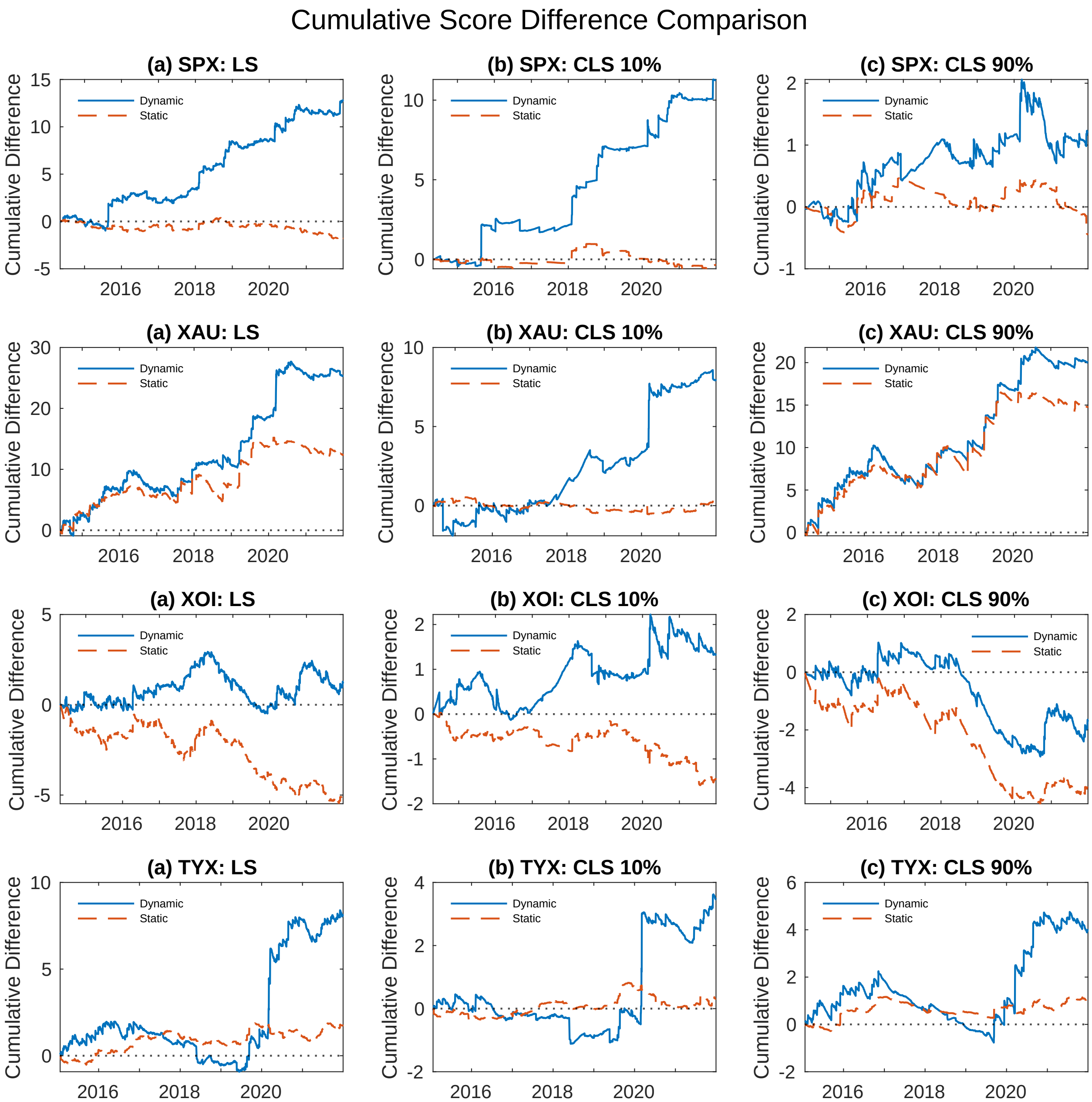
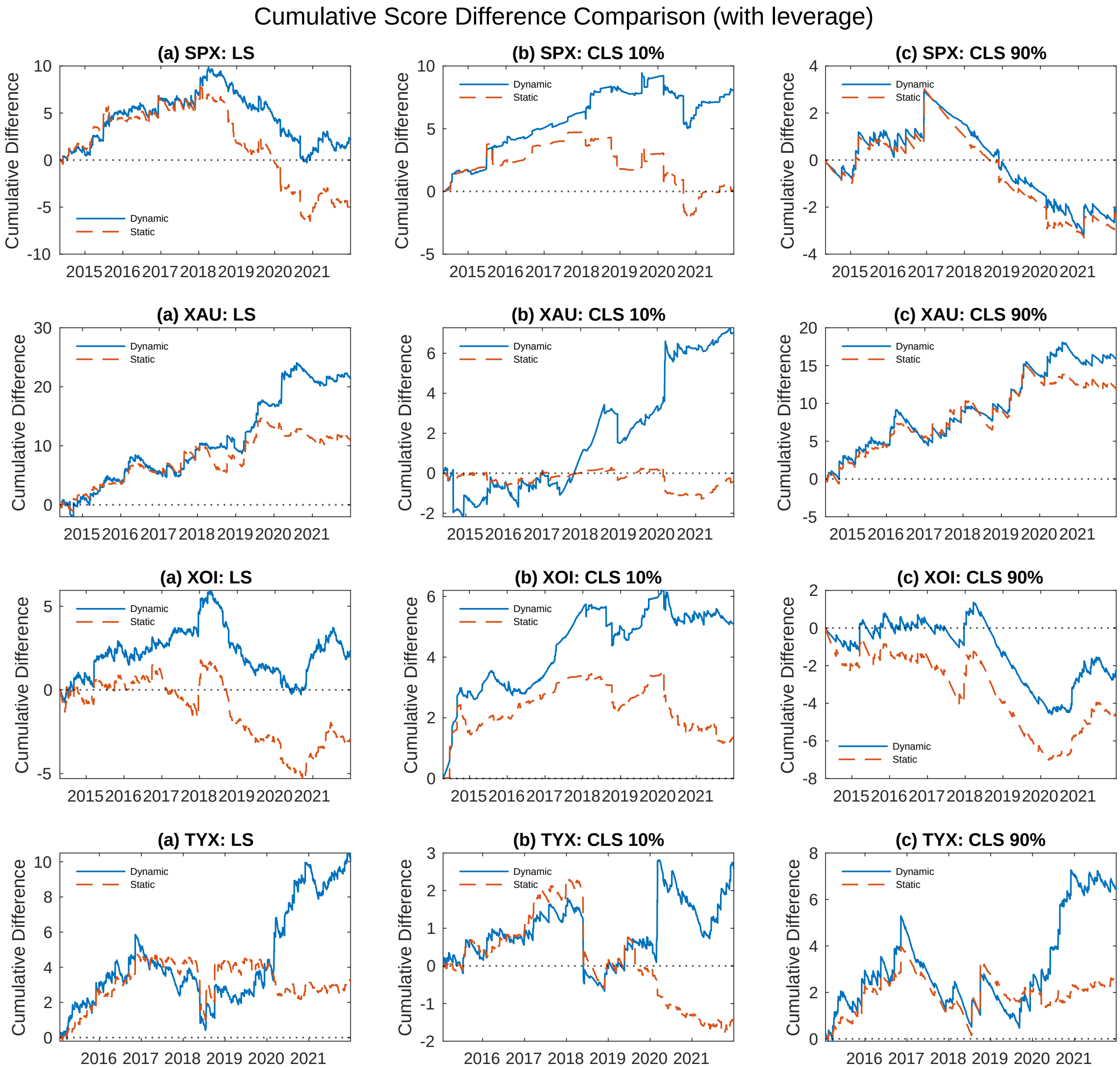


Figure 4: Cumulative score difference between DPM models and parametric SVJ model when leverage effect is included in the model. The blue solid line is the cumulative score difference between dynamic DPM and parametric SVJ model and the red dashed line shows the cumulative score difference between static DPM and parametric SVJ model. LS, CLS of 10% and 90% quantile are documented.



small gains from upper tail during that volatile time period. There are ups and downs of the cumulative LS difference, but overall, dynamic DPM provides better predictions than parametric model for SPX and XOI. Static DPM with leverage shows similar pattern for these two financial indices. For XAU and TYX, we obtain obvious gains from using dynamic DPM model. Not only for lower and upper 10% tails, but for the predictive density as a whole. During quiet periods, parametric SVJ model seems to work better than DPM models, resulting score difference around zero or being negative, such as 2015 to 2017 for XAU and 2018 to 2020 for TYX. However, during volatile periods, dynamic DPM provides much better predictive distribution for XAU and TYX, as we see a steep increase pattern around 2020. Static DPM model shows similar pattern, but does not provide as much gains as dynamic DPM during volatile periods and does not seem to differ from parametric model when predicting the 10% quantile of XAU.

5.4 Model Implied Jump Size Distribution

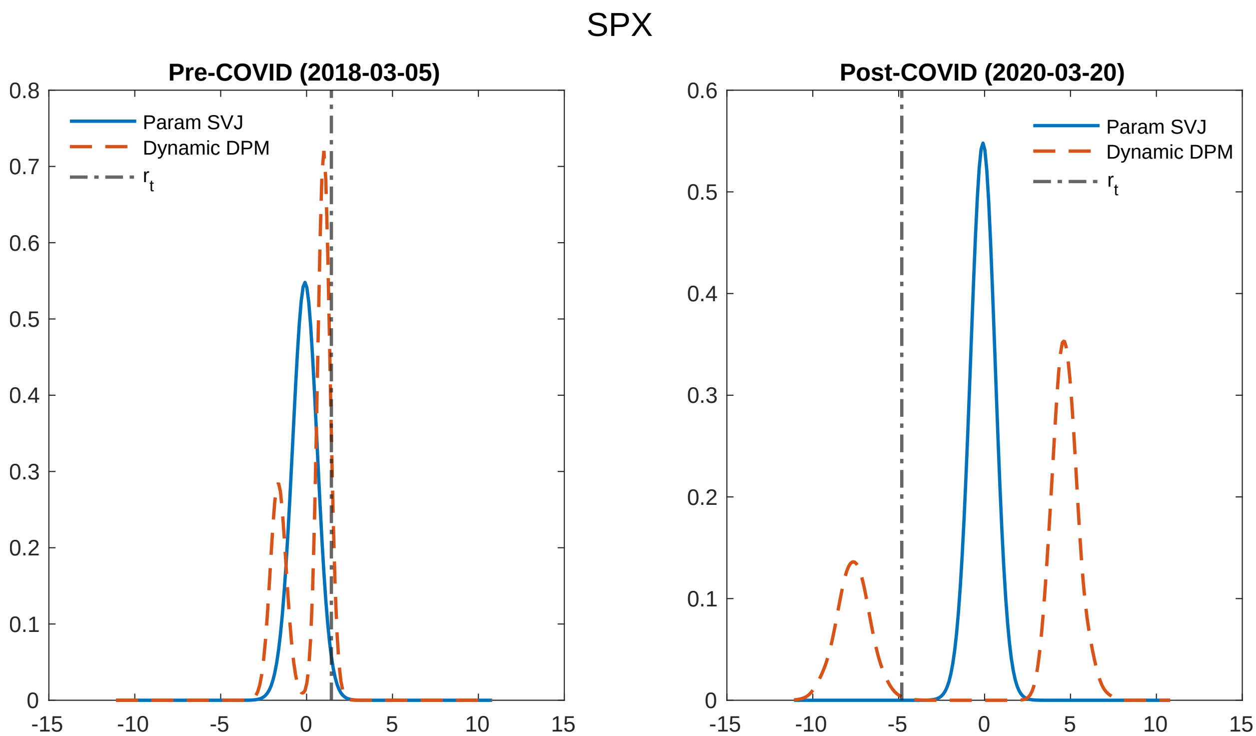
With the estimated parameters using the full dataset, the model implied jump size density at different time point t can be computed as:

$$\begin{aligned}
&\text{Dynamic DPM:}\\
&\qquad Z_t = \frac{1}{M}\sum_{i=1}^{M}\sum_{k=1}^{K^{(i)}} w_k^{(i)} N(\mu_k^{(i)} \bar{BV}_t^{1/2}, (\sigma_k^2)^{(i)})\\
&\text{Parametric SVJ:}\\
&\qquad Z_t = \frac{1}{M}\sum_{i=1}^{M} N(\mu_z^{(i)}, (\sigma_z^2)^{(i)}),
\end{aligned} \tag{30}$$

where $K^{(i)}$ is the selected number of groups based on slice sampler in i-th iteration for M iterations in total. With the dynamic DPM incorporating BV_t in the mean of jump size distribution, it allows the jump size density to vary over time, compared with the parametric SVJ model.

We compare the ex-post model-implied jump-size density for SPX before and after the onset of the 2020 pandemic. To illustrate how the distribution behaves under different market conditions, we select two representative trading days: 5 March 2018, a period of moderate market uncertainty (VIX = 18.73%) and $r_t = 1.4426\%$, and on 20 March 2020, during the height of the COVID-19 turmoil (VIX = 66.04%) and $r_t = -4.8200\%$. These dates are chosen because they represent different market conditions: when the return and volatility is moderate and when the return and volatility are both high. Figure 5 shows the model

Figure 5: The model implied jump size density for SPX before and after COVID. The estimated parameters are from the model without leverage effect. The jump size density from dynamic DPM, parametric SVJ model and the log return r_t on the day are plotted for comparison.



implied jump size density for SPX before COVID (left panel) and after COVID (right panel) for both dynamic DPM (the red dashed line) and the parametric SVJ model (the blue solid line). For pre-COVID time - a quiet day in the market, the coverage of both models overlaps a lot while the parametric SVJ model has a high mass around 0 and large variance but the dynamic DPM model has very low mass around 0 but higher mass around the extreme values. The observed return on that day (the black dot-dashed line) falls in the right mode of the dynamic DPM model where the parametric SVJ model has very small mass. For post-COVID time, the parametric SVJ model fails to adapt to the volatile market. The high mass around zero indicates that it doesn't successfully capture the jump size mean during a volatile period. Instead, the parametric SVJ model enlarges the variance and kurtosis to capture the extreme tails of the jump size density. However, the dynamic DPM model moves away from 0, and creates two modes around the extreme values. The observed return on the day falls in the left mode region of the dynamic DPM jump size density. Even though it does not fall in the high mass region, it is still covered by the left mode region. On the other hand, the parametric SVJ model has almost zero mass around the observed return value.

5.5 Application in Implied Volatility

The "smile/smirk" features observed from empirical option implied volatility is one of the motivations for the development of jump diffusion models. Correctly detecting jumps is useful for generating IV surface that fits better to the empirical one. Hence, we verify the practical use of our model with an implied volatility application. To construct the Monte Carlo simulation under risk-neutral measure Q, we make an extra assumption on constant jump intensity, in addition to the standard no-arbitrage assumptions of financial market.

With the estimated parameters, we assume r_t follows the process under risk-neutral measure Q as in following table: The risk premium is measured by $q = 0.01$ and $\mu_z^Q = 1.1\mu_z$. The value is carefully selected to ensure the $|\phi_1^Q| < 1$. The European call option prices and corresponding IV are then generated via Monte Carlo simulation using the estimated parameters in Table 6.

With jumps added to the SV models, it steepens the slope of IV and the average jump size can be proxied by the slope of IV smile (Eraker *et al.*, 2003; Eraker, 2004; Cont, 2010; Gatheral, 2011; Yan, 2011). As we focus on the usage of our jump model in identifying true jumps, we verify our model generated IV shape, specifically slope and curvature, with the empirical SPX call option IV.

As the jump component is included in the model, the IV expansion of the SVJ model in

Table 7: Model specifications under risk-neutral measure. $\mathcal{M}_1$ a parametric SVJ model with the jump size is assumed to follow a normal distribution. $\mathcal{M}_2$ is the dynamic SV-DPM jump model.

Models	Specification
$\mathcal{M}_1$: Parametric SVJ	$\begin{aligned} r_t^Q &= r_f + \exp(h_t^Q/2)\epsilon_t + \Delta N_t Z_t^Q \\ h_{t+1}^Q &= \phi_0 + (\phi_1 + q)h_t^Q + \sigma_h \eta_t \\ \begin{pmatrix}\epsilon_t \\ \eta_t\end{pmatrix} &\sim N\left(0, \begin{bmatrix}1 & \rho \\ \rho & 1\end{bmatrix}\right), \\ P(\Delta N_t = 1) &= \delta \\ Z_t^Q &\sim N(\mu_z^Q, \sigma_z^2) \end{aligned}$
$\mathcal{M}_2$: Dynamic SV-DPM jump model	$\begin{aligned} r_t^Q &= r_f + \exp(h_t^Q/2)\epsilon_t + \Delta N_t Z_t^Q \\ h_t^Q &= \phi_0 + (\phi_1 + q)h_{t-1}^Q + \sigma_h \eta_t \\ \begin{pmatrix}\epsilon_t \\ \eta_t\end{pmatrix} &\sim N\left(0, \begin{bmatrix}1 & \rho \\ \rho & 1\end{bmatrix}\right), \\ P(\Delta N_t = 1) &= \delta \\ Z_t^Q &\sim \sum_{k=1}^{K} w_k N(\mu_k^Q \bar{BV}_t^{1/2}, \sigma_k^2) \end{aligned}$

Aït-Sahalia *et al.* (2021) should be used. The $(1,2)$th order IV expansion is:

$$\begin{aligned}\Sigma^{(1,2)}(\tau, k, S_t) = \sigma_t^{(0,0)}(S_t) + \sigma_t^{(1,0)}(S_t)\epsilon + \sigma_t^{(0,1)}(S_t)k + \sigma_t^{(1,1)}(S_t)\epsilon k \\ + \sigma_t^{(-1,2)}(S_t)\epsilon^{-1}k^2 + \sigma_t^{(0,2)}(S_t)k^2 + \sigma_t^{(1,2)}(S_t)\epsilon k^2,\end{aligned} \tag{31}$$

where $\Sigma^{(i,j)}(\tau, k, S_t)$ is the (i,j)th order IV and S_t refers to the asset price, $\epsilon = \sqrt{\tau}$, $k = \log(K/S_t)$ is the log moneyness.

Table 8: The average difference of coefficients in eq.31 between the model generated IV and the empirical SPX IV. The bolded values are smaller difference to the empirical data coefficients. The SPX option data is ranged from 2016 to 2021.

IV Regression Coefficients							
	$\hat{\sigma}^{(0,0)}$	$\hat{\sigma}^{(1,0)}$	$\hat{\sigma}^{(0,1)}$	$\hat{\sigma}^{(1,1)}$	$\hat{\sigma}^{(-1,2)}$	$\hat{\sigma}^{(0,2)}$	$\hat{\sigma}^{(1,2)}$
Dynamic DPM	**0.0129**	-0.0681	**-0.5028**	**1.5334**	0.561	-2.1658	4.4326
Parametric SVJ	-0.0297	**-0.0143**	-0.5266	1.9213	**-0.3704**	**0.1577**	**3.3147**

Table 8 documents the average coefficient difference in eq.(31) between model generated IV and the empirical SPX IV. The empirical option data and predictive return and log volatility from the models between 2016 to 2021 are used to construct the above average coefficient difference. The bolded values are the smaller average difference, compared with the empirical SPX IV expansion coefficients. For $\hat{\sigma}^{(0,0)}$, $\hat{\sigma}^{(0,1)}$ and $\hat{\sigma}^{(1,1)}$, dynamic DPM are better than parametric SVJ. It represents that in terms of estimating the long term volatility, the slope on the log moneyness direction and the slope on the cross of log moneyness and time-to-maturity, the dynamic DPM fits to the empirical IV better. For the curvature of the log moneyness direction, the parametric model is better.

6 Conclusion

We propose a flexible jump model using DPM with two forms - static DPM jump model and dynamic DPM jump model. The difference between them is whether to incorporate high frequency volatility measure in the jump size distribution mean. With our simulation analysis under various settings, we show the dynamic DPM jump model can provide better jump identification than parametric SVJ model with normally distributed jump size and static DPM jump model. But the dynamic DPM jump model suffers from the small sample problem. When the jump intensity is extremely low, parametric SVJ model seems to be slightly better. Dynamic DPM jump model also generates a marginal posterior of jumps that fits the true jump density better. Moreover, we show that the dynamic DPM jump model can identify a bimodal posterior of jumps for four financial indices - SPX, TYX, XAU and XOI.

To further investigate the practical use of our model, we construct the predictive density of each index and find that in general, the dynamic DPM jump model generates better predictions for return distribution as a whole and for either lower or upper 10% quantile of the returns. We observe significant gains especially after very volatile period, such as 2020 pandemic. The model implied jump size density is also investigated for representative days corresponded to 2020 pandemic. We find that the dynamic DPM model implied jump size density is changing with time and volatility level and can cover the actual observed returns on the day. In addition, we verify our model using the option IV expansion. Compared with the parametric SVJ model, the IV generated from dynamic DPM model fits to the empirical IV curve better in the slope of log moneyness and the slope across the log moneyness and time-to-maturity. However, the parametric SVJ model is better if assessing the curvature across the log moneyness direction.

There are further development needed for the dynamic DPM model in this chapter. A more enhanced model set for comparisons, such as comparing dynamic DPM model with the parametric SVJ model with two jump processes, and further and more intensive investigation into the IV shapes will be completed in the future.

Moreover, there are several limitations of the proposed model that warrant further investigation. First, the model treats the jump events as independently and identically distributed, and therefore does not incorporate any jump event dynamics. In particular, the jump intensity is assumed to be constant over time, which is inconsistent with empirical evidence and may constrain the model's predictive performance. One possible extension is to adopt an infinite Markov switching framework, allowing both the jump intensity and the jump-size distribution to vary over time, with the latter governed by a hierarchical DPM. Second, the current specification does not accommodate volatility jumps or co-jumps between returns and volatility, features that are frequently observed in financial markets. Incorporating volatility jumps into the model is a natural avenue for future work.

References

Aıt-Sahalia, Y. (2004). Disentangling diffusion from jumps. *Journal of Financial Economics*, 74(3):487–528.

Aït-Sahalia, Y., Cacho-Diaz, J., and Laeven, R. J. (2015). Modeling financial contagion using mutually exciting jump processes. *Journal of Financial Economics*, 117(3):585–606.

Aït-Sahalia, Y., Li, C., and Li, C. X. (2021). Closed-form implied volatility surfaces for stochastic volatility models with jumps. *Journal of Econometrics*, 222(1):364–392.

Antoniak, C. E. (1974). Mixtures of Dirichlet processes with applications to Bayesian non-parametric problems. *The annals of statistics*, pages 1152–1174.

Barndorff-Nielsen, O. E. and Shephard, N. (2002). Econometric analysis of realized volatility and its use in estimating stochastic volatility models. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 64(2):253–280.

Barndorff-Nielsen, O. E. and Shephard, N. (2004). Power and bipower variation with stochastic volatility and jumps. *Journal of financial econometrics*, 2(1):1–37.

Barndorff-Nielsen, O. E. and Shephard, N. (2006). Econometrics of testing for jumps in financial economics using bipower variation. *Journal of financial Econometrics*, 4(1):1–30.

Bates, D. S. (1996). Jumps and stochastic volatility: Exchange rate processes implicit in deutsche mark options. *The Review of Financial Studies*, 9(1):69–107.

Bates, D. S. (2000). Post-'87 crash fears in the s&p 500 futures option market. *Journal of econometrics*, 94(1-2):181–238.

Black, F. (1976). Studies of stock price volatility changes. In *Proceedings from the American statistical association, business and economic statistics section*, page 177.

Black, F. and Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of political economy*, 81(3):637–654.

Chan, W. H. and Maheu, J. M. (2002). Conditional jump dynamics in stock market returns. *Journal of Business & Economic Statistics*, 20(3):377–389.

Christie, A. A. (1982). The stochastic behavior of common stock variances: Value, leverage and interest rate effects. *Journal of financial Economics*, 10(4):407–432.

Cont, R. (2001). Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative finance*, 1(2):223.

Cont, R. (2010). *Encyclopedia of quantitative finance*. Wiley.

Diks, C., Panchenko, V., and Van Dijk, D. (2011). Likelihood-based scoring rules for comparing density forecasts in tails. *Journal of Econometrics*, 163(2):215–230.

Duffie, D., Pan, J., and Singleton, K. (2000). Transform analysis and asset pricing for affine jump-diffusions. *Econometrica*, 68(6):1343–1376.

Eraker, B. (2004). Do stock prices and volatility jump? reconciling evidence from spot and option prices. *The Journal of finance*, 59(3):1367–1403.

Eraker, B., Johannes, M., and Polson, N. (2003). The impact of jumps in volatility and returns. *The Journal of Finance*, 58(3):1269–1300.

Escobar, M. D. and West, M. (1995). Bayesian density estimation and inference using mixtures. *Journal of the american statistical association*, 90(430):577–588.

Ferguson, T. S. (1973). A Bayesian analysis of some nonparametric problems. *The annals of statistics*, pages 209–230.

Fouque, J.-P., Papanicolaou, G., and Sircar, K. R. (2000). *Derivatives in financial markets with stochastic volatility*. Cambridge University Press.

Fulop, A., Li, J., and Yu, J. (2015). Self-exciting jumps, learning, and asset pricing implications. *The Review of Financial Studies*, 28(3):876–912.

Gatheral, J. (2011). *The volatility surface: a practitioner's guide*. John Wiley & Sons.

Geweke, J. and Amisano, G. (2010). Comparing and evaluating bayesian predictive distributions of asset returns. *International Journal of Forecasting*, 26(2):216–230.

Giacomini, R. and Komunjer, I. (2005). Evaluation and combination of conditional quantile forecasts. *Journal of Business & Economic Statistics*, 23(4):416–431.

Gneiting, T. and Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American statistical Association*, 102(477):359–378.

Gneiting, T. and Ranjan, R. (2011). Comparing density forecasts using threshold-and quantile-weighted scoring rules. *Journal of Business & Economic Statistics*, 29(3):411–422.

Heston, S. L. (1993). A closed-form solution for options with stochastic volatility with applications to bond and currency options. *The review of financial studies*, 6(2):327–343.

Hull, J. and White, A. (1987). The pricing of options on assets with stochastic volatilities. *The journal of finance*, 42(2):281–300.

Ishwaran, H. and James, L. F. (2001). Gibbs sampling methods for stick-breaking priors. *Journal of the American statistical Association*, 96(453):161–173.

Ishwaran, H. and James, L. F. (2002). Approximate dirichlet process computing in finite normal mixtures: smoothing and prior information. *Journal of Computational and Graphical statistics*, 11(3):508–532.

Jensen, M. J. and Maheu, J. M. (2010). Bayesian semiparametric stochastic volatility modeling. *Journal of Econometrics*, 157(2):306–316.

Kalli, M., Griffin, J. E., and Walker, S. G. (2011). Slice sampling mixture models. *Statistics and computing*, 21(1):93–105.

Kim, S., Shephard, N., and Chib, S. (1998). Stochastic volatility: likelihood inference and comparison with arch models. *The review of economic studies*, 65(3):361–393.

Kou, S. G. (2007). Jump-diffusion models for asset pricing in financial engineering. *Handbooks in operations research and management science*, 15:73–116.

Kou, S. G. and Wang, H. (2004). Option pricing under a double exponential jump diffusion model. *Management science*, 50(9):1178–1192.

MacEachern, S. N. (1994). Estimating normal means with a conjugate style dirichlet process prior. *Communications in Statistics-Simulation and Computation*, 23(3):727–741.

MacEachern, S. N. and Müller, P. (1998). Estimating mixture of dirichlet process models. *Journal of Computational and Graphical Statistics*, 7(2):223–238.

Maheu, J. M. and McCurdy, T. H. (2004). News arrival, jump dynamics, and volatility components for individual stock returns. *The Journal of Finance*, 59(2):755–793.

Maneesoonthorn, W., Forbes, C. S., and Martin, G. M. (2017). Inference on self-exciting jumps in prices and volatility using high-frequency measures. *Journal of Applied Econometrics*, 32(3):504–532.

Merton, R. C. (1976). Option pricing when underlying stock returns are discontinuous. *Journal of financial economics*, 3(1-2):125–144.

Neal, R. M. (2000). Markov chain sampling methods for dirichlet process mixture models. *Journal of computational and graphical statistics*, 9(2):249–265.

Neal, R. M. (2003). Slice sampling. *The annals of statistics*, 31(3):705–767.

Pan, J. (2002). The jump-risk premia implicit in options: Evidence from an integrated time-series study. *Journal of financial economics*, 63(1):3–50.

Scott, L. O. (1987). Option pricing when the variance changes randomly: Theory, estimation, and an application. *Journal of Financial and Quantitative analysis*, 22(4):419–438.

Sethuraman, J. (1994). A constructive definition of dirichlet priors. *Statistica sinica*, pages 639–650.

Walker, S. G. (2007). Sampling the dirichlet mixture model with slices. *Communications in Statistics—Simulation and Computation®*, 36(1):45–54.

Wiggins, J. B. (1987). Option values under stochastic volatility: Theory and empirical estimates. *Journal of financial economics*, 19(2):351–372.

Yan, S. (2011). Jump risk, stock returns, and slope of implied volatility smile. *Journal of Financial Economics*, 99(1):216–233.

Yu, J. (2005). On leverage in a stochastic volatility model. *Journal of Econometrics*, 127(2):165–178.

A Prior Specifications

Table 9: Prior specifications for parameters in simulation analysis

Simulation Analysis			
	Prior Spec.	Mean	Variance
SV model parameters			
(ϕ_0, ϕ_1)'	N($\boldsymbol{m_h = (0, 0.9)', H_h^{-1} = (1000, 1000)'}$)	(0,0.9)'	(1000,1000)'
σ_h^{-2}	Gamma($\frac{a_h}{2} = 2.01, \frac{b_h}{2} = 2.01$)	1	0.99
Jump model parameters			
δ	Beta($\alpha_0 = 2$,$\beta_0 = 5$)	0.2857	0.0255
$\boldsymbol{\mu}$	N($m = 0, h^{-1} = 1000$)	0	1000
$\boldsymbol{\sigma^{-2}}$	Gamma($\frac{a}{2} = 2.01, \frac{b}{2} = 2.01$)	1	0.99
$\boldsymbol{w}$	SBP($\alpha = 1$)	-	-

Table 10: Prior for empirical analysis

Prior distribution of static parameters		
SV model parameters:		
(ϕ_0, ϕ_1)	N($\boldsymbol{m_h, H_h^{-1}}$)	$\{\boldsymbol{m_h} = [0, 0.9], \boldsymbol{H_h} = [0.001, 0.001]\}$
σ_h^2	IG($a_h/2, b_h/2$)	$\{a_h = 5.01, b_h = 1.01\}$
Jump component parameters:		
δ	Beta(α_0,β_0)	$\{\alpha_0 = 10, \beta_0 = 130\}$
$\boldsymbol{w}$	SBP(α)	$\alpha = 1$
Parametric SVJ:		
$\boldsymbol{\mu}$	N(m, h^{-1})	$\{m = -3, h = 1\}$
$\boldsymbol{\sigma}^2$	IG(a/2,b/2)	$\{a = b = 2.01\}$
DPM Bimodal prior:		
$\boldsymbol{\mu}$	$p_\mu N(m_1, h^{-1}) + (1 - p_\mu)N(m_2, h^{-1})$	$\{m_1 = -3, m_2 = 3, h = 1, p_\mu = 0.5\}$
$\boldsymbol{\sigma}^2$	IG(a/2,b/2)	$\{a = 10.01, b = 3.01\}$

B Definition of High-frequency Measures

The quadratic variation at time t (QV_t) can be decomposed to the variation that comes from the continuous Brownian component - integrated volatility (IV_t) and the variation from the aggregation of squared jumps. It can be defined as:

$$
\begin{aligned}
QV_t &= \int_0^t \sigma_s^2 ds + \sum_{0<s\leq t} Z_s^2 \\
&= IV_t + \sum_{0<s\leq t} Z_s^2.
\end{aligned} \tag{32}
$$

Let $r_{t_i} = p_{t_{i+1}} - p_{t_i}$ be the ith of N observed returns between trading day t and $t+1$. Based on Barndorff-Nielsen and Shephard (2002) and Barndorff-Nielsen and Shephard (2004), RV_t works as a consistent estimator for QV_t on the t-th day, which is defined as:

$$
RV_t = \sum_{i=1}^{N} |r_{t_i}|^2 \xrightarrow{p} QV_t, \text{ as } N \to \infty. \tag{33}
$$

BV_t is defined by:

$$
BV_t = \frac{\pi}{2}(\frac{N}{N-1}) \sum_{i=2}^{N} |r_{t_i}||r_{t_{i+1}}| \xrightarrow{p} IV_t, \text{ as } N \to \infty, \tag{34}
$$

and then the difference between RV_t and BV_t can be used as a measure of jump variation,

$$
RV_t - BV_t \xrightarrow{p} \sum_{0<s\leq t} Z_s^2 \text{ as } N \to \infty. \tag{35}
$$